

Materials of the Earth–

What are materials of the Earth?

Materials of the earth are what can be seen as the building blocks of life.

These are the minerals, rocks, soil and water.

The outer most layer of the earth is the crust and it is responsible for majority of life on Earth.

The crust supports:

- The growth of plants
- The survival of animals
- The structure of our land and
- Development of human civilization

It is the combination of these materials that make life on Earth possible.

Matter

Matter consists of small units called elements. All elements that have been discovered are tabulated according to their various characteristics into the Periodic table.

The smallest mass of an element carrying its characteristics is an atom.

Atoms and Molecules

Atoms of the same kind and atoms of different kinds combine to form molecules.

Each molecule has the characteristics of the substance formed by such combination. For example sodium and chlorine combine to form common salt. (salt is a mineral). A grain of salt is a solid that has a specific shape and pattern.

What is a mineral?

When minerals are free to grow without constraint, they are bounded by crystal faces which are disposed in a regular way.

Minerals are thus naturally occurring inorganic solids that have a crystal structure and a definite chemical composition. Minerals are referred to as naturally occurring because they are formed through natural geological processes.

By crystal structure, we mean that the atoms that make up the mineral are arranged in an orderly manner in a 3D pattern that repeats itself. There are over 4500 different minerals identified on earth.

Physical properties of minerals

There is a close link between the structure of a mineral and its physical properties which are accordingly of considerable value in identification. These physical properties include:

- Hardness – resistance to scratch,
- Lustre – variation in brightness due to variation in reflecting surface
- Cleavage – tendency for a mineral to break along planes of weakness.
- Fracture – type of irregular breakage e.g. conchoidal in glass.
- Color – unique colors that vary in shade, intensity, and rate of tarnishing when exposed to air.

What are rocks?

Rocks are defined as naturally formed aggregates of minerals or mineral like substances.

Rocks can be made up of one type of mineral, several minerals or no minerals at all.

Limestone is a rock comprised of the mineral calcite. (mono-mineralic rock)

Many crystals of calcite are cemented together to form limestone.

Granite is a rock that is comprised of several different minerals including quartz, biotite, potassium, feldspar, and plagioclase.

Coal is an example of a rock that is not comprised of any minerals but instead is made of decomposed organic matter.

CLASSIFICATION OF ROCKS

Rocks are classified by how they were formed. The three main types of rocks are sedimentary, igneous and metamorphic.

Sedimentary rocks

The main sources of sediments are concentration of clastic materials (fragments of rock that have been removed from in-situ position). Sedimentation cycle incorporates separation of particles from the parent rock, transportation, deposition and the processes that lead to its consolidation into rock.

Clastic are formed by deposition of fragments or particles of rock and minerals. Deposition may be in-situ or in a different depositional environment.

Media for transport include water, ice, gravity etc. Deposition may be in lakes, flood plains, at deltas or in the open ocean. Since deposition generally takes place in quiet conditions, there is a high degree of sorting according to grain size. The shape of fragments will depend on distance travelled.

Rudaceous > 2mm pebbly

Arenaceous 0.06mm – 2 mm sands

Argillaceous < 0.06mm clay

Sediments are also derived from accumulation of organic matter either plants or animals, chemically derived sediments such as evaporites and pyroclastic type.

Sedimentation cycle incorporates separation of particles from the parent rock, transportation, deposition and the process that lead to its consolidation into rock.

Materials are buried, compacted, and cemented. The three processes are collectively referred to as lithification.

A combined process of lithification and chemical changes which take place are referred to as diagenesis. These processes give rise to a sedimentary rock.

Bedding, layering or stratification is a special characteristic of sedimentary rocks.

Examples of sedimentary rocks

#	Name of rock	Description
1.	Conglomerate	Made up of rounded pebbles cemented together
2.	Breccia	Made up of angular pebbles cemented together
3.	Sandstone	Sand grains cemented together into rock
4.	Shale	Made up of clay particles cemented together into rock
5.	Halite	evaporites
6.	Limestone	Shales of marine organisms
7.	Coal	Composed of organic matter from plant fragments

Igneous rocks

Igneous rocks are formed by the consolidation of molten or partially molten magma.

Origin

The magma appears to originate from the mantle through convectional movements.

Depending on the level of emplacement we end up with plutonic (formed beneath the surface), hypabyssal (minor intrusions such as dykes and sills – crystallized under conditions between plutonic and extrusive) or volcanic rocks

Magma starts rising as primitive magma.

As it rises, there is continual abstraction of heat resulting in some minerals being precipitated.

The precipitated ones are almost always more dense than the magma and sink and are removed from the magma.

From the initial starting composition, a great number of products can be obtained depending on the rate at which magma ascends relative to the rate at which heat is extracted from it.

The overall effect of differentiation is that successive residual liquids become enriched in silica and the alkalis sodium, and iron relative to magnesium.

In the early stages, large quantities of ferromagnesium minerals are removed. Further branching of the evolutionary line may occur depending on:

- Rate of movement
- Rate of heat flow loss
- Pressure conditions.

Thus basaltic magma plus silica enrichment results in [rhyolites](#) while those without silica enrichment results in [phonolite](#).

Characteristics of igneous rocks

The characteristics of igneous rocks are essentially the identifying features. These include colour, density, crystal type and textures. The factors affecting the texture are:

- (a) Viscosity of magma as it controls rate of ionic diffusion hence curtailing growth;
- (b) Cooling rate;
- (c) Order of crystallization of minerals;

(d) Relative rate of growth.

It is therefore possible to distinguish extrusive and intrusive rocks on the basis of texture.

Intrusives

These are intruded in deeper part of the crust and cools slowly to end up as medium to coarse grained. Well-formed crystals may be seen. Crystals are closely interlocking.

Extrusives

- Form near the surface;
- Cooling is rapid;
- Are fine grained to glassy; - examples – obsidian, basalt, trachyte, phonolite, ash, tuff, rhyolite.
- Spheroidal voids form in the rock as a result of gas trapped in the molten mass by the rapid cooling.

The extrusive may be lavas if the extrusion is peaceful and pyroclastic if the extrusion is violent.

Lavas

This are tabular igneous bodies thin compared to their horizontal extend.

Blocky lava (AA) - consists of irregular blocks.

Ropy lava – has smooth or ropy surface and occurs when volatile loss is less.

Pillow lava – rounded masses resembling pillows believed to be as a result of sub aqueous deposition.

Pyroclastics.

These rocks are accumulation of intensively fragmented rock from volcanoes of the explosive types. May include pre- existing igneous, metamorphic or sedimentary rocks.

They are usually associated with lavas of high silica content and high viscosity.

Unconsolidated products

Size	Unconsolidated products	Lithified products
33 mm+	Bombs/blocks	Agglomerate/volcanic breccia
4mm – 32mm	lapilli	Lapilli tuff
4mm	Ash and dust	Tuff.

Features of extrusive activities include composite and strato volcanoes (made up of alternating layer of lava and pyroclastic), shield volcanoes (not steep sided – less viscous hence slopes are gentle), lava flows/plateau/flood basalts

Volcanic activities give rise to rocks like – obsidian, basalt, trachyte, phonolite, tuff, ask, rhyolite.

Classification of igneous rocks

The rocks are classified and given names so that information about them can be conveyed in a condensed form.

Criteria: genetic, mineralogical or chemical

Igneous rocks grade into each other

A: Silica Content

This has a genetic implication as igneous rocks increase in silica content as they are differentiated.

Acid – more viscous melts, tend to form glass

- Less mobile, leads to steep volcanic cones
- More volatile and hence explosive

Basic- opposite of acid, less glass, more mobile, flatter volcanoes, less explosive.

Category	% silica	Rock name
Acid	>66%	Granite
Intermediate	52-66%	Andesite
Basic	45 – 52%	Gabbro
Ultrabasic	< 45%	Peridotite

B: Colour

Igneous rock forming minerals fall in two broad groups:

Felsic: low density, light in colour, relatively rich in silica, quartz and feldspars.

Mafic: high density, dark coloured, rich in iron and magnesium, includes pyroxenes, amphiboles, olivine and biotite

Ultramafic: igneous rocks containing virtually no quartz or felspar

C: Saturation concept

Igneous rocks are divided into those compatible with free quartz and those not with primary free quartz.

Those with primary free quartz – over saturated

Neither free quartz nor under saturated – saturated

Containing only unsaturated minerals - unsaturated

Observable characteristics of igneous rocks

1. Generally hard, massive, compact, with interlocking grains.
2. Absence of bedding plains
3. Usually contain feldspars
4. Crystals are generally interlocked

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METAMORPHIC ROCKS

These are formed by the alteration of pre-existing rocks by the action of temperature and pressure aided by fluids present.

Unlike sedimentary or extrusive igneous processes that may be observed at the surface, metamorphic processes take place at depth and are therefore not observable.

The pattern of a metamorphic rock is set by three variables.

- a) Nature of the original material
- b) The particular modifying process or processes
- c) The intensity with which these influences have operated

Textures differ from ones of igneous and clastic sediments chiefly in the way minerals fit together.

Metamorphic rocks for most part fit together without a paste and without interlocking.

The procedure in examining metamorphic rocks differ from the study of other types of rocks only in that their identification and interpretation call for close attention to the shape, spacing and orientation of minerals.

It is the custom to include data pertaining to texture in the single category of fabric.

The process of metamorphism may result in:

- Destruction of fossils, beds and sedimentary structures
- Hardening of rock
- Change of colour
- Alignment of minerals
- Growth of new metamorphic minerals

The visible changes brought about by metamorphism are:

- a) The partial or complete reorganization of the constituent minerals to form new mineral assembly
- b) Change in grain size, characteristically to produce coarse texture.
- c) The development of new structure particularly those manifested by parallel mineral arrangement

Metamorphic minerals

In general, the metamorphic minerals of greatest abundance are the same as those in igneous rocks and include: MICAS, CHLORITE, FELDSPARS, QUARTZ and HORBLLENDE

Stress minerals include: micas, albite feldspar, epidote, kyanite, staurolite, talc and garnet

Anti-stress minerals include: andalusite, cordierite, hypersthene, olivine and potassic feldspars

Summary of characteristics of the three types of rocks

Important features of metamorphic rocks

1. Generally hard, interlocking grains and bedded (if derived from stratified rocks)
2. Foliated, gneissose, schistose, slaty, granulose are the common structures.
3. Common minerals are andalusite, sillimanite, kyanite, cordierite, wollastonite, garnet graphite

Important features of igneous rocks

1. Generally hard, massive, compact with interlocking grains.
2. Absence of bedding plane
3. Usually contain much feldspars

Important features of sedimentary rocks

1. Generally soft, stratified i.e characteristically bedded
2. Fossils are common
3. Stratification, lamination, cross-bedding, ripple marks are usual features
4. Normally matrix is held together by a cementing material
5. Quartz, clay minerals, calcite, dolomite hematite are the common minerals.

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SOILS

Soils are mainly a product of weathering of rocks. It is a mixture of decomposed organic matter and broken down rocks and minerals.

This is as a result of weathering and erosion (both physical and chemical weathering).

On average soil is made up of 45 % rock and mineral pieces; 5% decomposed organic matter and 50% pore space.

The decomposed organic matter is responsible for producing nutrients for plants and increasing water retention.

The pore spaces in soil are very important because they allow water and air to circulate through the soil.

Water and air transport nutrients and carbon dioxide which are essential for the growth of plants.

